

NELVA ADALIT MORA BARRIONUEVO

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Systems Engineering student with interdisciplinary experience spanning software engineering, artificial intelligence, enterprise systems, business process optimization, and digital content strategy. Skilled in developing full-stack web applications, AI-powered solutions, e-commerce platforms, and business process models while leveraging data-driven decision making to improve organizational performance. Experienced as a Digital Content Creator, applying AI-assisted workflows, SEO, analytics, and omnichannel marketing to build online communities and digital products. Passionate about bridging technology, business, and innovation to solve real-world challenges.

EDUCATION

Universidad Mayor, Real y Pontificia de San Francisco Xavier de Chuquisaca

Candidate for Bachelor of Science in Systems Engineering (8th Semester) | Sucre, Bolivia (*Expected Graduation: 2027*)

Key Coursework: Software Engineering, Database Systems, Web Development, Machine Learning & Artificial Intelligence, Business Process Modeling (BPMN/Bizagi), System Dynamics & Simulation (Vensim), Operations Research (Linear Programming, CPM, PERT, Queueing Theory), Enterprise Systems (ERP Concepts & SAP Fundamentals), Organizational Design & Human Resource Management, Operations Management (MRP, Inventory Management & Forecasting), Economic Engineering (NPV, IRR, Cash Flow Analysis, Depreciation & Project Evaluation), Financial Statement Analysis, Electronic Commerce & e-Government, Scrum Master.

Ministerio de Educación de Bolivia

Technical Vocational Degree in Tourism and Hospitality (Técnico Medio) | Sucre, Bolivia (2022)

Acquired foundational training in client relations, logistics coordination, administrative management, and operations design.

PROJECTS & EXPERIENCE

Store LAN E-Commerce Platform | Lead Full-Stack Developer

Node.js, Express.js, PHP, Vanilla JS, CSS (Flex, Grid, Bootstrap), Supabase (PostgreSQL), JWT, Sharp, Render, Vercel

- **System Engineering & Deployment:** Designed and deployed a high-performance, responsive e-commerce web platform optimized for real-world transaction checkout.
- **Security Integration:** Developed and integrated a secure JWT token lifecycle handler on a decoupled RESTful API backend using Express and Node.js.
- **Database & Storage Optimization:** Integrated Supabase Database and Storage middleware, optimizing image processing with Sharp (Multer) to enhance page load speeds and asset delivery.
- **API Automation:** Automated a transactional coordinator flow that interfaces directly with WhatsApp APIs to confirm stock availability and coordinate QR payments.

GitHub Repository: <https://github.com/NelvaAdalit/Store-LAN.git>

Demo: <https://store-lan.vercel.app/>

Pediatric Pneumonia Computer-Aided Diagnosis | Deep Learning & Web Developer

Python, PyTorch, Vanilla JS, CSS3, Vercel, Render

- **Deep Learning Engineering:** Designed and trained a Multilayer Perceptron (MLP) Deep Learning classifier in PyTorch to detect pediatric pneumonia in chest X-ray images, achieving 94.8% diagnostic accuracy.
- **Backend Integration:** Developed a Python-based REST API backend deployed on Render to serve model inferences asynchronously.
- **Frontend UI/UX:** Built a clean, mobile-first responsive frontend UI deployed on Vercel to allow medical professionals to upload scans and inspect diagnoses in real time.

GitHub Repository: github.com/NelvaAdalit/Medical_Guangzhou_Pneumonia_Detection

Demo: <https://medical-guangzhou-pneumonia-detecti-omega.vercel.app/>

Digital Content Creator & AI Automation Specialist

Generative AI (LLMs, Diffusion Models), CapCut, InShot, Canva, SEO, Growth Marketing

- Produced multilingual educational and technology-focused content using Generative AI tools.
- Designed AI-assisted workflows for script writing, image generation, voice synthesis, and video editing.

- Applied SEO, audience analytics, and growth marketing strategies to build an organic community of over 11,000 followers, with video views peaking above 10,000 per post.
- Leveraged social media platforms to test digital marketing strategies and audience engagement.
- Managed end-to-end content production using AI, automation, and data analytics.

Account: https://www.tiktok.com/@evewheeler10?is_from_webapp=1&sender_device=pc

Demo: https://www.instagram.com/reel/DVYukmFCMEV/?utm_source=ig_web_button_share_sheet

Tabular Policy Gradient 8-Puzzle Solver | Machine Learning Developer

Python, PyTorch, NumPy, Pandas, Matplotlib, Jupyter Notebook

- **Reinforcement Learning:** Programmed a reinforcement learning agent using Tabular Policy Gradient methods to solve the classic 8-puzzle sliding grid.
- **Visualization & Metrics:** Serialized state-action policies into PyTorch weights (.pt format) and visualized learning curves to analyze agent optimization.

LEADERSHIP & ACADEMIC ACHIEVEMENTS

Aspire Leaders Program (Founded by Harvard University Faculty) | Global Cohort Fellow (2026)

Completed 40 academic hours of intensive global leadership training, emphasizing executive communication, ethical decision-making, and community impact.

Decanato de la Facultad de Ciencias y Tecnología (USFX) | Academic Juror (Jurado Evaluador) (2024)

Formally appointed by the Faculty Dean to act as a technical evaluator for Teaching Assistantship (Auxiliar de Docencia) selection exams, verifying candidates' academic integrity and competence.

NASA International Space Apps Challenge | Galactic Problem Solver Award & Sucre Finalist (2024)

Co-developed a software solution in a 48-hour global hackathon using open-source space agency data to address planetary challenges.

U.S. Embassy in Bolivia & Inspira Alumni | TechConnect Graduate & Selected Outstanding Project (2024)

Graduated from the intensive Technology & Entrepreneurship incubator; final project selected among the program's top entries.

Military Preparation & Training | Leadership & Resilience Formation

Completed rigorous physical and tactical military training, developing teamwork under pressure, crisis resilience, strict operational discipline, and task execution excellence.

TECHNICAL SKILLS & LANGUAGES

Programming: Python • JavaScript • PHP • SQL • HTML5 • CSS3

Software Development: Node.js • Express.js • PyTorch

AI & Machine Learning: PyTorch • NumPy • Pandas • Matplotlib • LLM Workflows

CMS & Digital Commerce: WordPress • WooCommerce • Elementor • Yoast SEO • Google Analytics • Joinchat

Business Systems & Operations: ERP Concepts • BPMN • Bizagi • Vensim • Operations Research • Process Modeling • Organizational Design

Business Analysis: Financial Analysis • NPV • IRR • Cash Flow Analysis • Depreciation • Project Evaluation • Inventory Management • MRP • Forecasting

Content & Design Tools: Canva • CapCut • InShot

Tools: Git • GitHub • Supabase • PostgreSQL • Render • Vercel • XAMPP

Methodologies & Systems: System Dynamics & Simulation (Vensim) • Business Process Modeling (BPMN / Bizagi) • Agile

Languages: Spanish (Native) • English: B1 (Working proficiency) • German: A1 (Elementary)